Given: • θ: skew shape

• $m \in \mathbb{Z}_+$

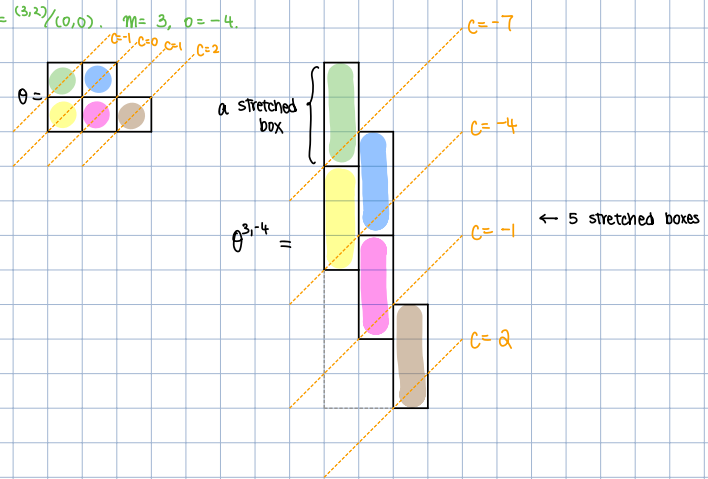
• $o \in \mathbb{Z}$

Construct a new skew shape $\theta^{m,o}$: For each box x of content c in θ, place m boxes of contents $o+mc,\ o+mc-1,\ \dots,\ o+mc-(m-1)$ in the same column as x.

The set of m boxes arising from x in this way is called a stretched box, denoted $\text{stretch}(x) \subseteq \theta^{m,o}$.

$\therefore$ The stretched boxes partition the boxes of $\theta^{m,o}$ into $|\theta|$ sets of size m.

e.g. $\theta = {}^{(3,2)}/_{(0,0)}$, $m=3$, $o=-4$.



Def 8.1.1: Given: • a tuple of skew shapes $\nu = (\nu_{(0)}, \dots, \nu_{(k)})$

• $m \in \mathbb{Z}_+$

• $\vec{o} = (o_1, \dots, o_k) \in \mathbb{Z}^k$ s.t. $0 \le o_1 \le \cdots \le o_k < m + o_1$

The associated m-stretching of ν is $\nu^m = \nu(m, \vec{o}) := (\nu_{(1)}^{m,o_1}, \dots, \nu_{(k)}^{m,o_k})$.

actually depends on $\vec{o}$.

* We use the same notation $\nu^m(i)$, $\nu^m(I)$ by treating ν^m as a tuple of skew shapes (independent of ν) for $I \subseteq [dm]$ where $|\nu^m| = dm$, $d = |\nu|$.

Let $\vec{A}(\nu)$ = set of attacking pairs $(\nu(i), \nu(j))$ in ν.

For $(\nu(i), \nu(j)) \in \vec{A}(\nu)$ where $\nu(i) \in \nu_{(r)}$, $\nu(j) \in \nu_{(s)}$, define $a_{ij} := |\{(a,b) \in \vec{A}(\nu^m) \mid a \in \text{stretch}(\nu(i)),\ b \in \text{stretch}(\nu(j))\}| = \begin{cases} m + o_r - o_s & \text{if } r < s \\ 1 + o_r - o_s & \text{if } r > s \end{cases}$.

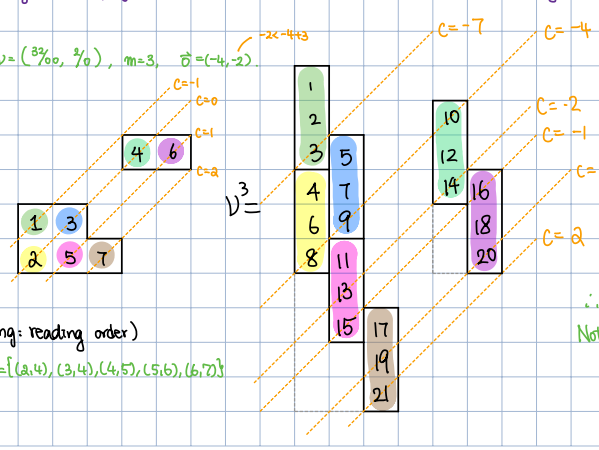
in the rth skew shape

in the sth skew shape

$\therefore$ $a_{ij} \ge 1$ and $a_{ij} - 1 = |\{(a,b) \in A(\nu^m) \mid a \in \text{stretch}(\nu(j)),\ b \in \text{stretch}(\nu(i))\}|$.

For $1 \le i < j \le d$, $(\nu(i), \nu(j)) \notin \vec{A}(\nu) \Rightarrow \{(a,b) \in \vec{A}(\nu^m) \mid a, b \in \text{stretch}(\nu(i)) \cup \text{stretch}(\nu(j))\} = \emptyset$.

e.g. $\nu = ({}^{32}/_{00}, {}^{2}/_{0})$, $m=3$, $\vec{o} = (-4, -2)$.



(filling: reading order)

$A(\nu) = \{(2,4), (3,4), (4,5), (5,6), (6,7)\}$

$a_{24} = |\{(8,10)\}| = 1 = 3 + (-4) - (-2)$

$a_{34} = |\{(9,10)\}| = 1 = 3 + (-4) - (-2)$

$a_{45} = |\{(10,11), (12,13), (14,15)\}| = 3 = 1 + (-2) - (-4)$

$a_{56} = |\{(15,16)\}| = 1 = 3 + (-4) - (-2)$

$a_{67} = |\{(16,17), (18,19), (20,21)\}| = 3 = 1 + (-2) - (-4)$

$m \quad + o_1 \quad - o_2$

$m + o_2 - o_1$

$\therefore$ $a_{24}, a_{34}, a_{45}, a_{56}, a_{67} \ge 1$.

Notice that $\{(a,b) \in \vec{A}(\nu^3) \mid a \in \text{stretch}(4),\ b \in \text{stretch}(2)\} = \emptyset$

same for $(4,3)$ and $(6,5)$

$\{(a,b) \in \vec{A}(\nu^3) \mid a \in \text{stretch}(5),\ b \in \text{stretch}(4)\} = \{(11,12), (13,14)\}$

$a_{45} = |\{(10,11), (12,13), (14,15)\}|$

Similarly for $a_{67} - 1$

Hence $a_{45} - 1 = |\{(a,b) \in \vec{A}(\nu^3) \mid a \in \text{stretch}(5),\ b \in \text{stretch}(4)\}|$.